The Antidote

Project overview

Models, data, and resilience

At a glance

The Antidote estimates when a ship will cross an analytical harbor gate in Hampton Roads. The historical replay compares quadratic regression, ordinary XGBoost, and Antidote's XGBoost model with conditional input repairs as AIS reports are delayed, missing, noisy, or biased. When an issue is detected, a human records its error type; those reviewed examples are saved and feed the next model-training cycle. The intended user is a port logistics coordinator deciding whether an arrival-dependent plan needs verification.

This overview pairs the full-March nominal comparison and repair experiments with the replay's March results. It explains how reports are disrupted and how the models respond.

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1 The problem

The intended user is a port logistics coordinator deciding whether to keep an arrival-dependent plan or verify it before committing resources. A late or unstable estimate can cause avoidable waiting, premature staffing, or repeated replanning. The prototype supplies an estimate and evidence for that decision; it does not allocate a berth, issue navigation instructions, or control a vessel. The historical replay supplies the interactive view of ships, incoming reports, data changes, and forecasts. Here, we explain the inputs and decisions behind those forecasts.

Our outcome is an *analytical harbor-gate crossing*, not docking, cargo delivery, or an official port-arrival timestamp. A qualifying east-to-west crossing intersects longitude $\lambda_g = -76.3^\circ$ between

latitudes 36.95° and 37.05°. Consecutive original observations must bracket it within 120 seconds. If the two longitudes are λ_0, λ_1, the reference is interpolated as

$$\alpha = \frac{\lambda_g - \lambda_0}{\lambda_1 - \lambda_0}, \qquad T^* = t_0 + \alpha(t_1 - t_0).$$

Interpolation assumes approximately linear motion inside the bracket. Sampling, position error, and uncertain destination intent remain limitations of the reference. The original bracket is fixed before introducing faults and never becomes a repair input. The comparison cohort includes approaches with an observed qualifying crossing.

We assess forecasts using MAE, MSE, 95th-percentile absolute error, prediction coverage, review burden, and condition-level results. Together, these measures show average and tail performance, forecast availability, and intervention needs.

1.1 The data and evaluation window

1.1.1 Source and target

The replay serves frozen Poly (A), XGBoost (B), and nominal-only Antidote (C) models across all 31 days of March 2024. Source and model hashes record the data lineage and configuration.

The March source is NOAA's cleaned broadcast-point AIS product, selected for Hampton Roads and filtered to cargo and tanker reports. The bundle records the March source checksum and model/code signatures. Its target is an interpolated analytical gate crossing, not verified docking or official arrival. NOAA's metadata describes collection and processing limits.[1]

Shared preprocessing selects cargo and tanker reports before simulated masking, parses UTC times, treats invalid navigation values as missing, and removes exact duplicate measurements consistently across models. MMSI identifies repeated vessel observations; it is not an independent observation count. Source archives already involve processing and incomplete receiver coverage, so "original" means unchanged by our simulator, not error-free ground truth.

1.1.2 Temporal split and information available at inference

The table below shows the fit, model-selection, and nominal-evaluation data for the replay models. An approach is one inbound vessel passage; a clock is one scheduled forecast during that passage. The partitions follow time order from 2023 training data to the March 2024 comparison.

Partition	Dates	Approaches	Clocks
Served model fit	1 Jan–29 Jun 2023	1,319	27,263
Served model selection	30 Jun–19 Jul 2023	140	3,108
Served model nominal evaluation	1–31 March 2024	243	5,382

[1] NOAA, Nationwide Automatic Identification System 2024.

The March comparison covers 219 vessels. Poly (A), XGBoost (B), and Antidote (C) are scored at the same scheduled clocks, so their forecasts can be compared as paired decisions.

A separate expanded-source retraining run uses 220 available NOAA daily archives from 1 January through 8 August 2023 (33,094,542 regional reports). It fits on 1 January–29 June (180 days), selects settings on 30 June–19 July (20 days), checks internally on 20–29 July (10 days), calibrates on 30 July–8 August (10 days), and evaluates the same March cohort of 5,382 scheduled decisions across 243 approaches and 219 vessels. The replay uses its frozen serving bundle; this retraining run is reported as a separate experiment.

Candidate evaluation protocol: fit on eligible 2023 source data; use a later chronological 2023 block for validation and model/repair selection; reserve a subsequent 2023 block for calibration; then freeze the model, repairs, and review thresholds before scoring the March 2024 cohort. Keep complete approaches together in each partition. Compare nominal, stale-only, low-quality-only, and combined conditions on matched cases and paired seeds. Report approach-weighted MAE and MSE, median and 95th-percentile absolute error, forecast coverage, review burden, and vessel-cluster uncertainty.

For report r_i, let t_i be its measurement time and d_i its simulated delivery time. At decision clock t, inference receives only

$$\mathcal{D}(t) = \{r_i : t_i \le t,\ d_i \le t\}, \qquad a(t) = t - \max_{r_i \in \mathcal{D}(t)} t_i.$$

The second quantity is the age of the latest measurement. Nominal delivery is instantaneous by assumption; it is not a measurement of actual receiver latency. Forecasts are scheduled every five minutes after at least ten minutes of an approach. Features summarize the ten minutes ending at the latest available observation, while age continues to increase at the decision clock. This preserves what an operator could know as time advances.

In the live replay, the reporting-gap control withholds reports for a chosen interval; it does not deliver old messages later. A separate saved stress curve tests delayed delivery. Neither substitutes for measured receiver latency.

2 The solution

2.1 Quadratic regression: a structured comparator

Model A expands the base features with degree-two polynomial terms and fits ridge regression. With feature expansion $\phi(x)$, it minimizes

$$\sum_i w_i \left(y_i - \beta^{\mathsf{T}} \phi(x_i)\right)^2 + \lambda \|\beta\|_2^2.$$

The base representation includes local east/north position, current speed, sine and cosine of course, recent mean speed and position, speed trend, history support, and explicit cargo/tanker/unknown-class flags. Circular course encoding avoids treating 359° and 1° as nearly a full rotation apart. Polynomial interactions provide a compact nonlinear comparator; regularization limits coefficient growth. This model is inspectable, but it can extrapolate poorly during waiting, turning, or unfamiliar approach patterns.

2.2 Ordinary XGBoost: a nonlinear comparator

Model B uses gradient-boosted regression trees on the same base feature contract and nominal training inputs. Predictions combine sequential tree contributions, $f(x) = \sum_{m=1}^{M} h_m(x)$. XGBoost supplies the nonlinear learner; it is the AI capability under evaluation, rather than merely a tool used to author the website.[2] Missing numeric features pass through an imputer fitted on training data. Unknown vessel class has its own flags and is not silently turned into cargo.

Both A and B predict remaining duration from the latest observation. Their absolute arrival prediction is $\widehat{T}_{A,B} = t_{\text{obs}} + f_{A,B}(x)$. Subtracting the current decision clock yields the remaining time now. An older observation can retain the same absolute forecast while its remaining duration shrinks; changing a display clock does not refresh the underlying evidence.

2.3 Antidote: degradation-aware learning and conditional repairs

The Antidote uses XGBoost. It adds robust motion summaries and observable freshness/quality signals: speed quantiles and variability, turn and course dispersion, a position-history velocity fit, report age, gap length, missing-field fractions, occupied minute bins, and disagreement between reported and position-derived motion. It predicts from the decision clock, $\widehat{T}_C = t + g(z)$. The model served in the current comparison is the nominal-only Antidote artifact: it was fitted on original reports, without injected deterioration. The separate augmented Antidote model remains experimental and is not served. Conditional repairs are added by the dashboard at inference time.

The trained guard limits report influence by forming one representative per UTC minute using medians and a circular course summary. It screens invalid coordinates and motion exceeding 60 knots across short gaps of at most 300 seconds. Minute binning limits the contribution of a prolific feed to history summaries; it does not make a biased majority within a minute trustworthy. Exact deduplication is shared with the ordinary comparators, so it cannot be credited exclusively to Antidote.

The dashboard adds conditional repair strategies to the Antidote. These are separate from its learned guard. The code keeps raw A/B inputs, original quality signals, support, labels, and clocks intact while changing only permitted derived inputs sent to the Antidote. Training uses squared-error regression; depth and learning rate are selected using approach-balanced development MAE. The configured search spans depths 2, 4, and 6, learning rates 0.03 and 0.1, up to 500 boosting rounds, and early stopping after 30 rounds. Choosing a model by MAE does not ensure its MSE or worst errors improve. The Kalman-based development approach tracks vessel motion sequentially: it predicts the next state from the prior estimate, then updates that estimate with each received AIS measurement. Its gain balances predicted-state uncertainty (process uncertainty) against sensor uncertainty (measurement uncertainty): noisier readings move the estimate less, while uncertainty that grows between reports lets new evidence move it more. The residual between the filtered estimate and the incoming reading can help identify persistent sensor drift, but it is not independent ground truth. The current main-branch implementation documented here still uses an EWMA persistence check and guarded, position-supported speed replacement; the Kalman approach is under development and is not part of these reported results.

[2] Chen and Guestrin, XGBoost: A Scalable Tree Boosting System, 2016.

2.4 Degradation scenarios

The replay studies accidental deterioration at *inference time*: a frozen model version receives worse reports. No live AIS broadcasts are altered, and a reviewed label does not silently change the model serving that replay. Instead, reviewed cases are retained for the next versioned training cycle. The replay injects controlled changes representing outages, unavailable telemetry, miscalibration, and navigation faults.

Fault	Controlled change	Mechanism affecting the forecast
Reporting gap	Withhold all reports in a fixed interval beginning ten minutes into an approach.	Old position and motion remain the latest evidence; data age increases while the ship continues moving.
Missing speed	Mask selected SOG values.	Current speed, recent summaries, and missingness evidence change. Zero speed remains a legitimate observation.
Missing course	Mask selected COG values.	Direction and turning features lose support; numerical imputation may follow a different tree path.
Missing vessel class	Mask selected type reports.	The model loses cargo/tanker context; unknown remains a distinct state.
Speed drift	Add a slowly varying bounded speed bias.	A persistent underestimate can resemble slower progress and shift predicted duration.
Location drift	Accumulate a bounded coordinate displacement.	Distance to the gate and position-derived motion become biased together.
Sensor deterioration	Add zero-mean Gaussian noise to SOG and COG.	Instantaneous errors propagate into direction, speed variability, and nonlinear tree decisions.
Position jump	Displace selected reports along a seeded direction.	Apparent progress or implausible motion changes position and quality features.

For illustration, the navigation perturbations can be written as

$$\widetilde{v}_i = \max\{0, v_i + b_v(t_i) + \epsilon_i\}, \qquad \widetilde{\mathbf{p}}_i = \mathbf{p}_i + \mathbf{b}_p(t_i) + J_i\mathbf{u}_i.$$

Here b_v is persistent speed bias, ϵ_i is random speed noise, $\mathbf{b}_p$ is coordinate drift, and $J_i\mathbf{u}_i$ is a position jump. The simulator also wraps noisy course into $[0, 360)^\circ$. Drift and noise differ: averaging can reduce zero-mean jitter under appropriate assumptions, while a persistent bias generally survives averaging.

The saved March graph comparisons use 25% of each control's configured maximum: a 15-minute reporting gap and 25% for each other control. The replay sliders affect live replay inputs independently; the graph selector uses the saved fixed-strength scenarios described in the results section.

Faults compose in a fixed order: gap, speed drift, location drift, noise, position jumps, and missing fields. Missing-field and position-jump selectors share a deterministic report hash and seed. Equal-severity selections therefore overlap subject to field eligibility; they are not independent dropout masks. Modification logs are constructed separately. Audit labels, original altered values, and future crossing outcomes are exposed to the audience only after inference.

2.5 Repairs, warnings, and deferral

2.5.1 Reconstructing missing motion and noisy sensors

When positions are sufficiently consistent, a short local track can estimate motion. For local east/north position $\mathbf{p}_i$ and times centred around $\bar{t}$, the least-squares velocity is

$$\widehat{\mathbf{v}} = \frac{\sum_i (t_i - \bar{t})(\mathbf{p}_i - \bar{\mathbf{p}})}{\sum_i (t_i - \bar{t})^2}, \qquad \widehat{s} = \frac{\|\widehat{\mathbf{v}}\|}{0.514444}, \qquad \widehat{\theta} = \operatorname{atan2}(\widehat{v}_E, \widehat{v}_N).$$

Positions are in metres and time is in seconds; $\widehat{s}$ is therefore in knots. Course is converted to degrees and wrapped around the circle. A stable trajectory can support estimates for missing speed and course independently. Both use original received positions; one reconstructed sensor does not become evidence for the other. Each historical correction uses only information available when that particular report arrived, preventing later reports from backfilling an earlier estimate.

The Gaussian-noise strategy checks a five-minute track with at least three distinct reports, at least 60 seconds of span, no gap above 180 seconds, and fit residual at most 50 metres. Additional speed, turn, and plausibility checks reject unsuitable references. It replaces a supported reading when its speed differs by more than 2 knots or its moving-course difference exceeds 15°. This is *guarded replacement*, not a fitted blend of the reported and estimated readings. A smoother track is still only a conditional reference, not independent ground truth.

2.5.2 Detecting persistent bias

Speed drift is checked against position-supported speed. For disagreement $r_k = v_k - \widehat{s}_k$, the detector forms a time-aware exponentially weighted estimate,

$$e_k = (1 - \rho_k) e_{k-1} + \rho_k r_k, \qquad \rho_k = 1 - \exp(-\Delta t_k / \tau), \qquad \tau = 2 \text{ minutes}.$$

The provisional rule requires same-direction disagreement across at least three reports and two minutes, with $|e_k| > 0.5$ knots. Reference acceleration, track fit, and turning checks can prevent reconstruction. These conditions reduce impulsive replacements but also reject legitimate manoeuvres; the screen finds limited repair availability for speed drift.

Location drift compares observed displacement with motion implied by received SOG/COG. Persistent displacement disagreement can justify verification, but it cannot identify the true coordinates by itself. The dashboard therefore warns rather than correcting location. A separate experimental mode projects from explicit, time-available position anchors for at most five minutes and requires trusted navigation. Those anchors are simulated from original AIS in the experiment, so they are an oracle-reference study, not an independent operational feed. Coherent corruption of both position and motion may evade these consistency checks.

2.5.3 Fallbacks, metadata, and combined faults

The reporting-gap strategy detects silence from observed cadence and does not fabricate reports. Position-jump detection supplies navigation-consistency warnings and does not repair coordinates. Missing class can be carried forward only from consistent recent received metadata for the same MMSI; conflicts remain unknown and require review.

Position faults pause position-based navigation repairs. Noise plus speed drift pauses their overlapping sensor corrections; class recovery can continue independently. The current fleet policy selects permitted reference assumptions from the simulated fault families enabled in the controls. It is not blind automatic fault isolation. Pausing avoids asserting an unsupported correction, but it is not evidence that the forecast has recovered. A production system would need independently justified reference trust and an evaluated fallback policy.

3 The results

3.1 Replay graph readout

The four replay graphs summarize March scenario results across the model comparisons and selected data faults. Each selected fault uses the fixed 25% setting. Seeded variation and interpolation expand the saved checkpoints into smooth curves. The impact view reports absolute ETA error, forecast movement shows how estimates change over time, and arrival error compares the forecast with the recorded crossing.

With no faults selected, the replay graph summaries are:

Graph measure (min)	Poly (A)	XGBoost (B)	The Antidote (C)
Impact view: absolute ETA MAE	6.20	4.32	4.10
Forecast movement at 30 minutes	10.96	5.49	3.87
Arrival error at 60 minutes	14.21	10.21	6.92
Arrival error at crossing	0.58	0.38	0.59

Table 3: Replay graph results with no faults selected.

The daily ETA error graph shows signed error through March. Forecast movement is the mean absolute shift from the initial ETA; arrival error is mean absolute ETA error at the indicated time before the recorded crossing. Selecting faults updates the displayed results.

Full-March nominal baseline. The serving bundle scores Poly (A), XGBoost (B), and the nominal-only Antidote (C) model on the same 5,382 scheduled decisions across 243 approaches and 219 vessels. These are original, unmodified March reports; no injected faults or additional dashboard repairs are applied in this comparison. MAE is averaged within each approach, then across approaches. MdAE and P95 are pooled across individual forecast errors. The Antidote's MAE is 0.033 minutes (about 2 seconds) below XGBoost's. The paired vessel-bootstrap 95% interval for Antidote minus XGBoost is $[-0.125, +0.050]$ minutes and spans zero. Poly (A) has the lowest P95 error. These are point predictions before any interval-based deferral.

Model	Approach-weighted MAE (min)	Forecast MdAE (min)	Forecast P95 AE (min)
A: quadratic ridge	7.019	3.459	36.725
B: ordinary XGBoost	6.643	3.164	37.361
Antidote (C): nominal-only XGBoost	6.610	3.097	37.097

3.2 Staleness

At each decision clock, the model receives reports only after their simulated delivery time, and report age continues increasing until a newer measurement arrives. The live replay's gap control withholds reports, while a separate saved stress curve recomputes forecasts under delayed delivery; these are distinct scenarios. Nominal delivery is assumed instantaneous, so neither is a measurement of real receiver latency. The baseline table above gives the full-March reference; stress graphs show how forecasts move as delayed evidence accumulates.

3.3 Low-quality data and source volume

The replay separately exposes missing fields, noisy or drifting sensors, and position jumps, then allows these conditions to be combined. Fault definitions and affected fields are shown above. Fault strength and source volume are different questions: strength changes how unreliable an affected report is, while source volume changes how many reports come from that source. A fixed-volume prevalence test varies the fraction of unreliable reports while holding the total report count constant. Repetition tests ask whether repeated reports from one source gain influence; total-volume tests ask whether adding reports changes that influence when the unreliable share stays fixed. Exact duplicates are removed by shared preprocessing for every model, so only nonidentical repetition can test whether a source gains weight through additional observations.

Volume matters when downstream summaries give a more prolific source more influence. Here, minute-level medians and bounded history summaries limit that influence, but a biased majority within a time bin can still move its representative value.

3.4 Over-time behavior

The ordered replay advances through received reports and forecast updates. Saved comparisons cover all 256 on/off subsets of the eight controls beginning at the March 15 cutover. The graphs do not recalculate when live sliders move. Their saved predictions have source/model/code signatures and 108 scalar forecast checks against the serving path.

3.5 Countermeasure effectiveness and limits

The live view shows Antidote's conditional repair, the affected report fields, and whether the available evidence supports correction or requires review. Separate same-case experiments compare the frozen optimized Antidote model (C) with each repair disabled and enabled on identical schedules and original crossing labels. At 25% simulated report masking on the April 8–21 follow-up cohort, the saved results are:

Condition	Antidote MAE	Antidote + repair	Change [95% interval]	Coverage
Missing speed	6.2740	5.9377	−0.3363 [−0.4250, −0.2421]	100%
Missing course	6.1865	5.9265	−0.2600 [−0.4130, −0.0835]	100%
Missing vessel class	5.8755	5.8598	−0.0157 [−0.0312, −0.0038]	100%

Table 4: Approach-weighted ETA MAE in minutes. Negative paired change favors the repair.

The three studies cover 112 approaches, 110 vessels, and 2,402 scheduled clocks; partial masks use three paired seeds and uncertainty intervals resample vessels. All forecasts remain in the coverage denominator, including clocks where the repair is unsupported. These are executed, versioned experiments.

Evidence note: These measurements use simulated faults on the April follow-up cohort and show same-case effects for the three tested repairs. Sensor-noise ETA results use the legacy v1 model; location correction uses an additional simulated trusted-position channel; position-jump handling is warning-only. Reproduction reports are in the `results/REPORT.md` files under `missing_speed_mitigation`, `missing_course_mitigation`, and `missing_vessel_class_mitigation`.

Other controls have complementary evidence. Under speed-only drift, the policy lowers speed-reading MAE from 2.6326 to 2.3313 knots (11.4%), repairs 13.11% of decision-time readings, and requests review for the rest. With an additional simulated trusted-position stream, location correction lowers position disagreement from 1,015.90 to 93.69 metres (90.8%); its 0.9341-minute ETA improvement was measured on legacy C v1. Sensor-noise replacement on legacy C v1 lowers ETA MAE from 9.5404 to 6.7999 minutes at high noise, and from 20.5170 to 7.1345 minutes at extreme noise. At 25% corruption, the position-jump monitor flags 97.71% of assessed affected pairs and warns on 100% of affected windows.

The single-fault objectives build on the repair comparisons above; the broader objectives extend the comparison across clean, stale, low-quality, and combined conditions.

Single-fault objectives: target about 5.4% lower ETA MAE for missing speed, 4.2% for missing course, and 0.3% for missing vessel class, each versus its own attacked Antidote baseline. Retain the observed 100% forecast coverage. These percentages describe separate conditions and should not be added together.

Broader one-month objectives:

- **Nominal:** reduce approach-weighted MAE by 12% versus current Model B, from 6.643 to about 5.846 minutes, while reducing MSE by at least 10% and retaining 100% coverage.
- **Staleness:** when 25% of reports are withheld for ten minutes, reduce MAE by at least 20% versus the same frozen XGBoost model without repair, with at least 95% coverage.
- **Source volume:** when nonidentical reports from an unreliable source are doubled, keep MAE within 10% of nominal and improve MAE by at least 20% versus the same stressed model without repair.
- **Combined faults:** across the eight controls at 25% strength, lower MAE by at least 15% and MSE by at least 10% versus the same attacked cases with repairs disabled, retain at least 95% coverage, and limit clean-feed MAE regression to 2%.
- **Across time and feedback:** reproduce the direction of the combined-fault gain across all 31 March days and three paired seeds; after one review-informed training cycle, lower MAE by at least 10% on later repeat-fault cases, cut false-positive review requests on clean reports by 20%, and retain at least 95% sensitivity to injected faults.

Together, the full-March baseline, eight-control replay, and repair studies connect clean performance with examples of data deterioration and correction.

Across the product, the reference is an observed analytical harbor crossing. The replay applies defined fault scenarios to compare model behavior.

For approach j with n_j available forecasts and signed error $e_{jk} = \widehat{T}_{jk} - T_j^*$ in minutes, the primary accuracy metrics are

$$\mathrm{MAE} = \frac{1}{J}\sum_{j=1}^{J}\frac{1}{n_j}\sum_{k}|e_{jk}|, \qquad \mathrm{MSE} = \frac{1}{J}\sum_{j=1}^{J}\frac{1}{n_j}\sum_{k}e_{jk}^2.$$

Coverage uses all scheduled cases as its denominator, including unavailable forecasts. Large-error percentiles pool clocks. Paired uncertainty should resample vessels, retain repeated approaches within a vessel, and be reported alongside condition-level outcomes. The current full-March nominal comparison uses a paired vessel bootstrap.

4 Where humans are involved

When Antidote detects a data-quality issue or an estimate that warrants attention, the reviewer sees the received timestamp, affected fields, diagnostic evidence, and available forecasts. The reviewer identifies the error type when the evidence supports one (for example, a stale report, missing field, sensor noise, speed drift, location drift, or position jump); uncertain cases remain unresolved rather than being forced into a category. The reviewed label and its evidence are saved as a feedback record for later model training. The reviewer does not overwrite the original AIS observation or invent an arrival outcome.

For a reviewed event i, let x_i be the evidence available at the decision time and let z_i be its human-adjudicated error type. The feedback store retains the event context and model version alongside

(x_i, z_i). A later training cycle adds these reviewed examples to the versioned training set, so a candidate can learn which observable patterns correspond to recurring error types:

$$\widehat{\phi}_{v+1} = \arg\min_{\phi} \left[-\sum_{i \in \mathcal{F}_v} \log p_\phi(z_i \mid x_i) \right].$$

Here $\mathcal{F}_v$ contains only reviewed feedback available to training version $v+1$. If a verified crossing outcome y_i is also available, its example can contribute to the ETA objective, $\ell_{\mathrm{ETA}}(f_\theta(x_i), y_i)$; an error-type label by itself is not treated as ETA ground truth. This creates a repeatable improvement loop: detect, review, save, retrain, and validate. New candidates are evaluated as versioned models before they replace the frozen replay model, so improvement is the goal of each cycle rather than a guaranteed outcome of every retrain.

An operator should inspect the received timestamp, altered-field evidence, active warning, and available estimates before accepting an unreliable arrival-dependent plan. Simulator audit highlighting explains the experiment and is not used as a human error label or a deployable fault detector. A position inconsistency calls for checking an independent navigation or port source. Antidote exposes conditional estimates without overwriting the displayed received measurements; correction of a source report belongs to the accountable data provider or a separately verified input workflow.

The latest status policy separates an unavailable optional noise/drift reconstruction from an actionable review request. Turning or accelerating can invalidate a simple track fit without proving a fault. Such a case shows "Estimate unavailable" while preserving the input evidence. Detected inconsistencies, blocked corrections, missing required support, and other existing review warnings remain. This follows the general principle that alerts should prompt a useful human action.[3]

Human-reviewed error types and their supporting context are saved for retraining. These labels provide feedback about observed failure modes, but do not by themselves measure alert precision or operator workload.

5 Data governance and limitations

The reviewed NOAA product is publicly accessible; its metadata lists coastal and ocean planning as the use constraint. Public access alone is not blanket authorization for redistribution or operational navigation. Retain attribution, preserve the selected product's metadata, and verify applicable terms before publishing raw subsets or acquiring a live feed. The serving bundle includes model artifacts, a March replay cache, and hashes; source training CSVs are not included in the tracked bundle. The 2023 acquisition record still needs source URLs, retrieval dates, geographic filters, completeness notes, and counts tied to the exact training view.

MMSIs and precise historical tracks identify vessels and can expose operational patterns. We use identity only for grouping and display, without attempting to identify crew, enrich operator profiles, or join unrelated personal information. A live pilot would need explicit access, retention, logging, and sharing rules. The evaluation covers Hampton Roads cargo and tanker approaches with observed crossings. Accuracy can be compared by vessel class, forecast horizon, data support, and previously seen vessels.

[3] Google SRE, Monitoring Distributed Systems.

The feedback records retain only the information needed to interpret and reproduce a reviewed event: the relevant input snapshot, assigned error type, model version, review state, and a verified crossing outcome when available. Simulator-generated fault flags remain experiment metadata and do not become human labels. In a proposed pilot, the data steward owns provenance, access, retention, and permitted use; the model lead owns versioned retraining, validation, drift monitoring, and rollback; an operations reviewer owns adjudication and escalation. The submitting team remains accountable for claims and source review. AI assistance was used for implementation and drafting. A language model is not part of the ETA inference path.

6 Next steps

The present decision is **iterate**. The product works as a historical replay, and the nominal March comparison has a saved results and verification record. Next work focuses on expanding the fault scenarios and completing the all-2023 training protocol.

1. **Model lead:** finish the all-eligible-2023 training protocol using time-ordered model selection within 2023. Freeze model and repair choices before the final March scoring pass. Using the existing offline CPU workflow and matching authorized source files, evaluate every March clock, multiple fault seeds, isolated and combined conditions, coverage, MSE, tail errors, and the same-C repair ablation. Repeat source-volume controls and investigate any large-error cases using development data only; no live AIS integration is needed for this study.

2. **Data steward:** complete the 2023 and March acquisition manifests, including source URLs, dates, geographic filters, checksums, rights, and retained approach counts. Verify redistribution terms and obtain an independent position reference before making coordinate-recovery claims.

3. **Data integration:** connect timestamp-aligned weather observations and other relevant data sources with AIS reports, document their provenance and quality checks, and evaluate how they contribute to ETA forecasts.

4. **Geographic expansion:** extend the evaluation from Hampton Roads to additional ports, define comparable analytical crossing points at each port, and evaluate forecast performance across locations.

The checkpoint is a versioned report showing nominal and degraded accuracy, MAE/MSE and large-error behavior, output coverage, alert misses, review burden, and resource use. Revisit a controlled pilot only if the reach goals hold without unacceptable tail-error or workload tradeoffs, a port partner approves data access, and accountable roles and non-actuating boundaries are in place. Stop or narrow the scope if those conditions fail.

For a ten-minute judge walkthrough, spend one minute on the port-planning decision and crossing reference, three minutes on the clean replay and nominal comparison, three minutes showing a reporting gap and a sensor-quality fault, two minutes on Antidote's repair and a case where it cannot safely correct, then one minute on the main takeaways.

7 Reproducibility

The current nominal comparison, model artifacts, March replay cache, hashes, split counts, and inference signatures are saved under `Data_Science/artifacts/data_eval/optimized/` and `Data_Science/experiments/nominal_model_comparison/`. Its results are reported above as the current full-March nominal comparison. The attack graphs are saved in `Frontend/tracking/data/scenarios.jso` their metadata records the selected scenarios, fixed strength, source/model/code hashes, and scalar parity checks.

The serving bundle contains the model artifacts needed for replay; the website does not train on startup. Full setup and preparation commands are in the repository's frontend guide and model guide. From `Data_Science`, the normal service command is:

```
python -m Optimized_Model serve --port 8765
```

The saved March comparison bundle can be checked with `python experiments/static_graphs/export.py -verify-only`. Its source/model/code signatures and scalar forecast checks support consistency with the serving path, not operational correctness.

1. NOAA Office for Coastal Management. Nationwide Automatic Identification System 2024. Data lineage, fields, constraints, and receiver-coverage limits.
2. U.S. Coast Guard Navigation Center. How AIS Works. Reported position, motion, and vessel-information context.
3. Chen, T., and Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. Nonlinear learner used in XGBoost (B) and Antidote (C).
4. Google SRE. Monitoring Distributed Systems. Actionable alerting and excessive-interruption risks.